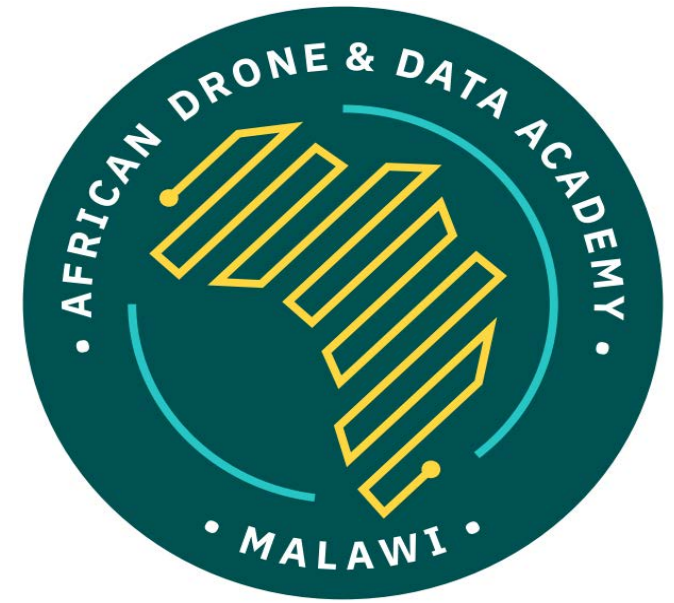




TOP Level 1

Day 5 Version A





TOP Level 1

Evidence of Licenses, Certificates and Approvals



Required Certifications

- The operator of an RPA is required to possess appropriate licenses, certifications or permits to operate an RPA commercially

1. Evidence of licenses, certs, approvals
2. Organization
3. Responsible person
4. Insurance
5. Airmanship
6. Code of conduct

DCA RPA regulations state:

56. — (1) No person shall engage in commercial, corporate and non-profit operations unless the operator holds a valid RPA Operator Certificate (ROC) issued by the Authority.

(2) The ROC shall authorise the operator to conduct RPA operations in accordance with the conditions and limitations detailed in the operations specifications attached to the ROC.

(3) The issuance of an ROC by the Authority is dependent upon the RPA operator demonstrating an adequate organisation, method of control and supervision of flight operations, training programme as well as ground handling and maintenance arrangements consistent with the nature and extent of the operations specified and commensurate with the size, structure and complexity of the organisation.

(4) A commercial RPA operator shall establish and implement a safety management system (SMS) in accordance with their operational requirements.



TOP Level 1 Organization

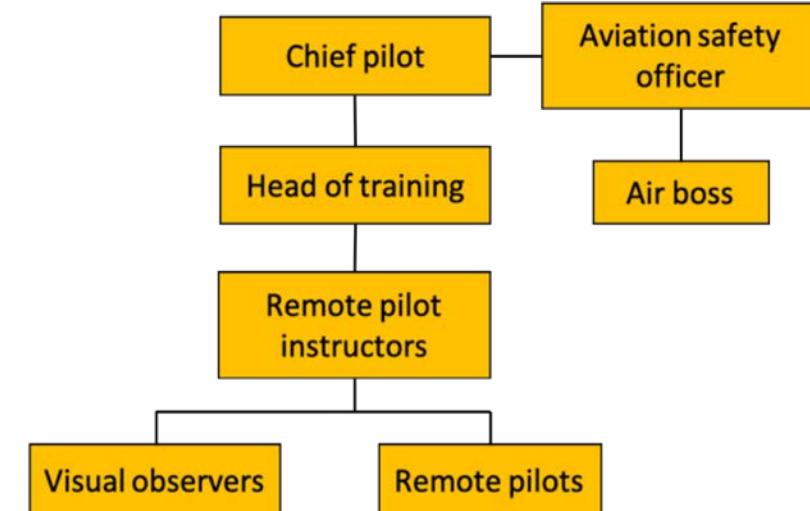


Organizational/management structure

- The ADDA has in place an organization that allows oversight for safe flight practice
 - The **Chief Pilot** is the key accountable person for all flight operations and training, and has TOP level 2 certification
 - The **Head of Training** ensures curricular goals and objectives are met, and has TOP level 2 certification
 - The **Aviation Safety Officer (ASO)** oversees safe flight operations through reference to the Safety Management System (SMS) rulebook. The ASO has TOP level 1 certification and has passed the visual observer (VO) knowledge exam

1. Evidence of licenses, certs, approvals
2. **Organization**
3. Responsible person
4. Insurance
5. Airmanship
6. Code of conduct

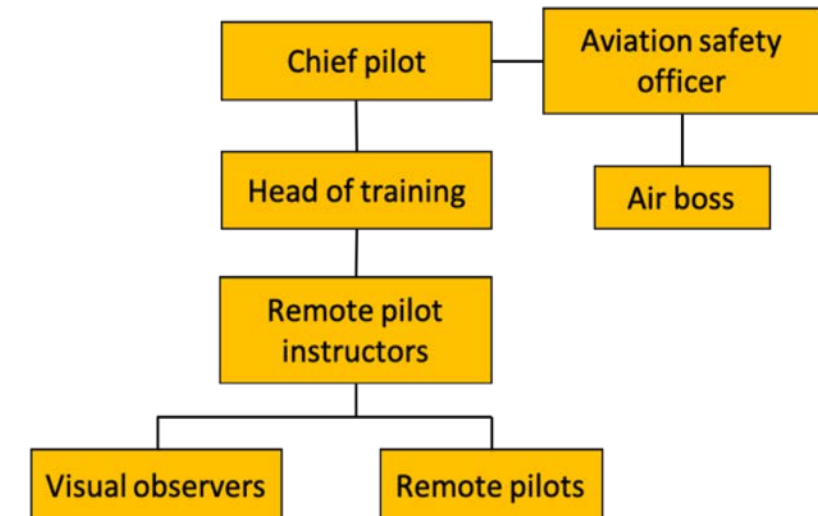
See FOM Sec. 2.0



Organizational/management structure

- The ADDA has in place an organization that allows oversight for safe flight practice
 - The **Remote Pilot Instructor (RPI)** delivers the training curriculum and tracks student performance. The RPI possesses a remote pilot license (RPL) and TOP level 2 RPI certification
 - **Remote Pilots** are either Student Pilots (SP) or Remote Pilots in Command (RPIC). The SP operate under privileges in the Flight Operations Manual (FOM) § 5.1.1
 - **Visual Observers (VO)** are responsible for communicating operational hazards and air traffic conflicts during flight activities

1. Evidence of licenses, certs, approvals
2. **Organization**
3. Responsible person
4. Insurance
5. Airmanship
6. Code of conduct





TOP Level 1

Responsible Person



The Responsible Person for all flight operations is the **Remote Pilot in Command (RPIC)**

- If a flight activity is occurring, there will always be a Remote Pilot in Command who is the single point responsible person for that flight
- The RPIC is responsible for:
 - Safe flight operations
 - Altering a flight operation to maintain a minimum level of safety in the event that anomalous behavior is detected
 - The RPIC is the final authority to make and/or approve of all operational decisions in a flight activity

1. Evidence of licenses, certs, approvals
2. Organization
3. **Responsible person**
4. Insurance
5. Airmanship
6. Code of conduct





TOP Level 1

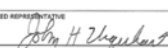
Insurcance



Insurance requirements

- 9.** 1) A person shall not operate, or cause to be operated or commit any other person to operate RPA unless there is in force an adequate insurance policy in respect of third party risks.
- 2) The minimum sum of insurance in respect of any RPA insured in accordance with sub-regulation (1) shall be notified to the Authority.
- 3) An operator of RPA shall make available third party liability insurance certificate(s), in the authentic form, at the location of the RPA operator's operational management or other location specified by the Authority.

1. Evidence of licenses, certs, approvals
2. Organization
3. Responsible person
4. **Insurance**
5. Airmanship
6. Code of conduct

 ACORD CERTIFICATE OF LIABILITY INSURANCE		DATE (MM/DD/YYYY) 5/20/2017
THIS CERTIFICATE IS ISSUED AS A MATTER OF INFORMATION ONLY AND CONFERS NO RIGHTS UPON THE CERTIFICATE HOLDER. THIS CERTIFICATE DOES NOT AFFIRMATIVE OR NEGATIVELY AMEND, EXTEND OR ALTER THE COVERAGE AFFORDED BY THE POLICIES BELOW. THIS CERTIFICATE OF LIABILITY INSURANCE DOES NOT CONSTITUTE A CONTRACT BETWEEN THE SUBROGATEE(S), AUTHORIZED REPRESENTATIVE OR PRODUCER, AND THE CERTIFICATE HOLDER.		
IMPORTANT: If the certificate holder is an ADDITIONAL INSURED, the policy(ies) must be endorsed, if SUBROGATION is waived, subject to the terms and conditions of the policy, certain policies may require an endorsement. A statement on this certificate does not confer rights to the certificate holder in lieu of such endorsements(s).		
PRODUCER Risk Management Plan of the Commonwealth of Virginia - Self Insured	CONTACT NAME: John Ushakov PHONE: 540-231-7433 FAX: 540-231-0064 E-MAIL: johnu4@vpi.edu	INSURANCE AUTHORIZED AGENT'S COVERAGE INSURER A: State Insurance Reserve Fund Trust created in INSURER B: Section 2.5-526.5 of the Code of Virginia INSURER C: State Property Risk Management Plan created in INSURER D: Section 2.1-523.3 of the Code of Virginia INSURER E: INSURER F:
INSURED Commonwealth of Virginia and all of its agencies, including Virginia Tech, Mail Code 0310 Blacksburg VA 24061		
COVERAGES CERTIFICATE NUMBER REVISION NUMBER		
THIS IS TO CERTIFY THAT THE POLICIES OF INSURANCE LISTED BELOW HAVE BEEN ISSUED TO THE INSURED NAMED ABOVE FOR THE POLICY PERIOD EXCLUDED, NOTWITHSTANDING ANY RIGHTS, TERM OR CONDITION OF ANY CONTRACT OR OTHER DOCUMENT WITH RESPECT TO WHICH THIS CERTIFICATE MAY BE ISSUED OR MAY PERTAIN. THE INSURANCE AFFORDED BY THE POLICIES DESCRIBED HEREIN IS SUBJECT TO ALL THE TERMS, EXCLUSIONS AND CONDITIONS OF EACH POLICY. LIMITS SHOWN MAY HAVE BEEN PROVIDED BY THIRD PARTIES.		
TYPE	TYPE OF INSURANCE	POLICY NUMBER
A COMM. GENERAL LIABILITY ☐ CLAIMS MADE ☒ OCCUR COMM. AUTOMOBILE LIABILITY PER: POLICY ☐ BIPOL ☐ LPPOL ☐ LPPOL	See Additional Remarks Sheet	07/01/2016 05/20/2016 EACH OCCURRENCE \$ 2,000,000 BODILY INJURY AND PROPERTY DAMAGE \$ MED EXP INJURY \$ PERSONAL & ADV INJURY \$ GENERAL AGGRIEVANCE \$ PRODUCTS / COMPOUND \$ COVERAGE SINGLE UNIT \$ BODILY INJURY PER PERSON \$ BODILY INJURY PER INCIDENT \$ PROPERTY DAMAGE \$ ADV INJURY \$ EACH OCCURRENCE \$ AGGRIEVANCE \$ PER PERSON / PER INCIDENT \$ E.L. BACK ACCIDENT \$ E.L. STRESS IN EMPLOYEE \$ E.L. DISEASE - POLICY UNIT \$
C Property Insurance	See Additional Remarks Sheet	07/01/2016 05/20/2016 See Additional Remarks Schedule
DESCRIPTION OF OPERATIONS / LOCATIONS / VEHICLES (Check Accord 19). Additional Remarks Schedule, if more space is required. Activities of Virginia Tech employees and volunteers, while acting within the scope of their duties and responsibilities as representatives of Virginia Tech. All-Risk Property Insurance covering fire and theft.		
CERTIFICATE HOLDER DuPont State-Hazlett Research Center 1900 Elkins Road Newark DE 19711		CANCELLATION SHOULD ANY OF THE ABOVE DESCRIBED POLICIES BE CANCELLED BEFORE THE EXPIRATION DATE THEREON, NOTICE WILL BE DELIVERED IN ACCORDANCE WITH THE POLICY PROVISIONS. AUTHORIZED REPRESENTATIVE 
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TOP Level 1

Airmanship



Airmanship principles

- Know your aircraft, the capabilities and limitations in normal and non-normal scenarios.
- Know yourself, your limitations and personal habits that effect safety.
- Know your environment, the airspace, the weather, the operation.
- Know your team, conduct briefings, use your observers, communicate, be positive and resolve conflicts quickly and fairly.
- Know the Risks, always conduct a thorough risk assessment.
- Stay Proficient, practice and conduct 'dry runs' of complex operations, and scenario based training for emergencies.
- Have the discipline to keep up to date on the latest safety and operational knowledge, including technical information regarding your platform.
- Do this to improve your situational awareness on every operation

1. Evidence of licenses, certs, approvals
2. Organization
3. Responsible person
4. Insurance
- 5. Airmanship**
6. Code of conduct

Know your aircraft, the capabilities and limitations in normal and non-normal scenarios.

- A review of the Standard Operating Procedures Manual (SOPM) will provide the operator with *aircraft specific* guidelines on safe flight practice
 - This will include operating limitations, emergency procedures, and other flight modes that the operator should be aware of

S-500 Operating limitations:

- *Flight in any type of precipitation is prohibited.*
- *Surface wind limit – 15 knots.*
- *Winds aloft limit – 15 knots.*
- *The flight must be conducted in VFR conditions (surface visibility must be 1 SM and the cloud ceiling must be at least 1,000 ft.).*
- *The S500 must remain within visual line of sight (VLOS) of the PIC.*
- *Flights must occur within the altitude specifications set form in this document.*

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6. Code of conduct

Know yourself, your limitations and personal habits that effect safety.

- Conduct a self-assessment of your readiness for flight, using the IMSAFE checklist:
 - **I** = Illness: Do I have any symptoms?
 - **M** = Medications: Have I been taking any prescription or over-the-counter drugs that might affect my performance
 - **S** = Stress: Am I under psychological pressure from the job? Am I worried about financial matters, health problems, or family discord?
 - **A** = Alcohol: Have I been drinking within the past 8 hours?
 - **F** = Fatigue: Am I tired and not adequately rested?
 - **E** = Emotion: Am I emotionally upset?

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6. Code of conduct

Know your environment, the airspace, the weather, the operation.

- Prior to flight, you have completed a flight planning process described in the FOM, Section 3.5:
 - You have verified that flight visibility allows visual line-of-sight of the aircraft
 - You have verified that clouds are above your flight altitude
 - You have conducted a site assessment, verified that the airspace is clear and you have computed risk that shows an acceptable level of risk

1. Evidence of licenses, certs, approvals
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6. Code of conduct

Risk Likelihood	Risk Severity			
	Catastrophic	Critical	Moderate	Minor
	D	C	B	A
4 – Likely	4D	4C	4B	4A
3 – Occasional	3D	3C	3B	3A
2 – Seldom	2D	2C	2B	2A
1 – Improbable	1D	1C	1B	1A

Know your team, conduct briefings, use your observers, communicate, be positive and resolve conflicts quickly and fairly.

1. Evidence of licenses, certs, approvals
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6. Code of conduct

- All pilots must complete the standard briefing checklist (AC01) prior to flight which allows the RPIC to conduct a briefing with the flight crew
- Crew Resource Management (CRM), Sterile Cockpit and standard terminology are reviewed prior to flight
 - **See Section 3.5.7 of the FOM**

Note that human error is a contributing factor in 70 - 80% of accidents in civil and military aviation



Know the Risks, always conduct a thorough risk assessment

- Overall risk is calculated from a weighted series of risks.

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Risk Likelihood	Risk Severity			
	Catastrophic	Critical	Moderate	Minor
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2 – Seldom	2D	2C	2B	2A
1 – Improbable	1D	1C	1B	1A

Stay Proficient, practice and conduct 'dry runs' of complex operations, and scenario based training for emergencies

1. Evidence of licenses, certs, approvals
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4. Insurance
5. **Airmanship**
6. Code of conduct

- Always practice difficult maneuvers using the flight simulator before proceeding to the actual flight operations
- Review ground control commands and settings, and verify with another pilot before executing
- Review communications terminology with crew (Section 3.5.7.4 FOM)
- Review emergency response in the SMS



Have the discipline to keep up to date on the latest safety and operational knowledge, including technical information regarding your platform

1. Evidence of licenses, certs, approvals
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6. Code of conduct

Company websites have FAQ or technical support links which should be the first line of inquiry to solve a particular problem

Use a software tracking log to find the latest software that is being used on equipment - review it frequently



SUPPORT FOR PHANTOM 4 PRO V2.0

FAQ



Aircraft



Gimbal and Camera



Remote Controller



Video Downlink



Battery

1. How can I tell the Phantom 4 Pro V2.0 and Phantom 4 Pro apart?

2. Are the ESCs used in the Phantom 4 Pro V2.0 the same as those in the Phantom 4 Pro?

Improve situational awareness for every operation

- Each crewmember displays the proper concern for balancing safety with operation accomplishment.
- Illustrating the value of review, feedback, and critique focusing on the process and the people involved.
- One of the best techniques for reinforcing effective human factors practices is careful debriefing of activities, highlighting the processes that were followed.
- It is essential that the crew be able to recognize good and bad communications, improvable techniques, and effective and ineffective team behavior.
- Crews handle errors and disagreements in a professional manner, avoiding personal attacks or defensive posturing.

1. Evidence of licenses, certs, approvals
2. Organization
3. Responsible person
4. Insurance
5. **Airmanship**
6. Code of conduct



TOP Level 1

Code of Conduct



The AUVSI UAS Code of Conduct presents best practices for RPA pilots

1. Evidence of licenses, certs, approvals
2. Organization
3. Responsible person
4. Insurance
5. Airmanship
6. **Code of conduct**

I pledge to:

- Follow airmanship principles, contained in the TOP PCM.
- Maintain the appropriate licensing, training, proficiency and competency on the UAS system being used.
- Conduct UAS flights only after a thorough assessment of risks associated with the activity. Risk assessment elements are detailed in the TOP PCM, at a minimum, they will include:
 - Check that weather and wind conditions support safe operations relative to the performance capability of the system.
 - Know my system's limitations and prepare for contingencies (lost link, power plant failures, loss of control, etc.). I will have an emergency plan ready.

The AUVSI UAS Code of Conduct presents best practices for RPA pilots

- ***I pledge to*** conduct UAS flights only after a thorough assessment of risks associated with the activity.
- Brief myself, my crew and spectators on the flight details and safety issues and protect my control area.
- Confirm that my crew and I are fit for flight operations and free from the effects of drugs, alcohol, illness or fatigue.
- Comply with aviation regulations as appropriate to the operation and airspace. I will monitor the airspace during all flight operations for potential conflicts.
- Establish a relationship and communicate with local law enforcement and public safety officials as appropriate. Know how to reach them in an emergency.
- Confirm that I am familiar with equipment reliability, performance, and airworthiness standards, that all my paperwork is in order and at hand, including insurance and aircraft manuals.

1. Evidence of licenses, certs, approvals
2. Organization
3. Responsible person
4. Insurance
5. Airmanship
6. **Code of conduct**

The AUVSI UAS Code of Conduct presents best practices for RPA pilots

- ***I pledge to*** conduct UAS flights only after a thorough assessment of risks associated with the activity.
- Ensure all my equipment is working and compliant: Communication, command, control and payload frequency spectrum requirements. I will develop contingency plans.
- Comply with all federal, state, and local laws, ordinances, covenants, and restrictions, including privacy, nuisance and trespassing.
- Be respectful and responsive to the needs of the public and environment.
- Cooperate fully with federal, state, and local authorities in response to emergency deployments and mishap investigations.
- Promote integration into the national airspace by following regulations and utilizing LAANC and other approved tools.

1. Evidence of licenses, certs, approvals
2. Organization
3. Responsible person
4. Insurance
5. Airmanship
6. **Code of conduct**

The AUVSI UAS Code of Conduct presents best practices for RPA pilots

1. Evidence of licenses, certs, approvals
2. Organization
3. Responsible person
4. Insurance
5. Airmanship
6. **Code of conduct**

I pledge to:

- Share my experience to improve safety performance for others in the industry.
- Encourage other remote pilots to follow the AUVSI UAS Code of Conduct and Safety Culture Pledge.



TOP Level 1

Safety Management System



The Safety Management System (SMS) is established to promote a safety culture within the ADDA

7. SOPs
- 8. SMS**
9. Work safety
10. Crew roles

- The SMS outlines RPA safety procedures
- Ensures proper UAS safety procedures are adhered to by all ADDA-affiliated students and personnel
- Creates a culture of safety within the operations and among the people
- Ensures the proper documentation and recording of all flight safety procedures and incident reports
- Ensures safety related matters are properly communicated through the personnel framework of the ADDA

Safety committees are appointed to maintain a high safety culture standard

7. SOPs
8. **SMS**
9. Work safety
10. Crew roles

- The **Internal Safety Committee (ISC)** has personnel that oversee safety matters within the ADDA
 - Committee members are comprised of the Aviation Safety Officer, Project Manager, Head of Training, Chief Pilot, and a Local/Site representative
 - Safe operations are enforced and safety risks are mitigated by:
 - Forming and maintaining a comprehensive and beneficial safety culture;
 - Establishing organizational priorities and tasking;
 - Outlining procedures on how to perform activities or processes;
 - Ensuring accountability to all established safety procedures;
 - Maximizing the skills of the organization's personnel;
 - Allocating all necessary resources to address safety concerns, and above all
 - Demonstrating proper safety procedures in all operations.

Safety committees are appointed to maintain a high safety culture standard

7. SOPs
8. **SMS**
9. Work safety
10. Crew roles

- The **External Safety Review Committee (ESRC)** is comprised of an external panel of RPA experts that oversee ADDA operations from an external perspective
 - This committee represents an outside body that holds the ADDA accountable for its safety culture

The SMS defines several areas of activity to support a safety culture

7. SOPs
8. **SMS**
9. Work safety
10. Crew roles

- Incident and accident investigation: the ISC conducts investigations with outcomes reported to the ESRC
- Emergency response plan (ERP): The ERP defines a response to an emergency
- Occupational health and safety (OHS): Policies are established to protect the well-being of the flight crew (pilots, visual observers and other essential personnel)

The SMS defines several areas of activity to support a safety culture

7. SOPs
8. **SMS**
9. Work safety
10. Crew roles

- **Occupational Health and Safety** addresses areas of concern to pilot and crew performance, including:
 - Crew fatigue
 - Protection from the elements
 - Dehydration and heat related risks
 - Sun
 - Thunderstorms
 - Personal protective equipment (PPE)
 - Distractions and interference

The SMS defines several areas of activity to support a safety culture

7. SOPs
8. **SMS**
9. Work safety
10. Crew roles

- **Hazard safety zones** are defined to keep operators at a minimum safe distance away from aircraft during launch and recovery
 - VTOL aircraft have a minimum safe distance of **1m**: a distance that should not be reduced at any time during the launch and recovery operation
- **General operational guidance** is provided by the Aviation Safety Officer who will determine the acceptability of new areas of operation
 - The RPIC and Chief Pilot review all flight requests and provide concurrent acceptance to initiate the flight operation